

Northeastern University

Uncovering MBTA Ridership Trends Through Data

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DS2500 Intermediate Programming with Data

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Problem Statement and Background

Public transportation systems like the Massachusetts Bay Transportation Authority (MBTA) are essential to life in Boston, providing access to jobs, education, and community resources. As students, many of us rely on the MBTA to get to classes, co-ops, and many other opportunities on a daily basis. However, we often don't have a clear understanding of the ridership trends that drive service decisions such as timing and frequency of routes. In the post-pandemic era, these trends have shifted significantly, and understanding these changes is crucial for ensuring that transit services remain efficient and responsive to our needs.

While MBTA analysts work with ridership data internally, as students, we have limited insight into how these patterns actually shape our daily commutes and the transit system as a whole. Our goal is to gain a better understanding of these trends, particularly as they relate to different times of day, days of the week, service lines, etc. This project is a learning opportunity for us to better comprehend how the MBTA operates, and to think critically about how the transit system impacts us and our peers.

We chose this project because MBTA ridership directly affects our lives, and understanding its patterns is key to navigating the transit system more effectively. By exploring data on broader ridership trends, we hope to uncover insights that can help us become more informed riders, even if the scope of our analysis is modest. Overall, this project reflects our desire to make sense of how the MBTA works and become more informed students, especially since the MBTA is a frequent topic of discussion given its significant impact on our daily lives. Through our analysis, we hope to understand better the trends that influence our daily commutes and feel more empowered as we navigate the city's public transit system.

Introduction to Our Data

We acquired ridership data from the [MBTA Data Portal](#). The two datasets we used listed in the portal under the tag ‘ridership’ are [MBTA Monthly Ridership By Mode and Line](#) and [Fall 2024 MBTA Rail Ridership by Hour, Route/Line, and Stop](#). The former shows ridership averages and totals over time, separated by numerous factors including service line and day of week. The latter includes boarding totals and flow for each hour at each stop across all train service lines throughout Fall 2024, accounting for the direction the train is traveling.

For data collection, the MBTA utilizes two main sources: automated passenger counters (APCs) and the automated fare collection (AFC) system. APCs use infrared beams and other technologies to count all riders who enter vehicles, while the AFC system counts CharlieCard and ticket taps. For services without significant APC coverage, estimates are used based on data from the AFC system, adjusted to account for riders who do not interact with the fare box and for transfers that happen behind the fare gates. On the commuter rail, rides are manually counted at various times and places to check against automatically collected data. An important point to note is that to measure ridership, the MBTA counts each time a rider boards a vehicle. Individual rides are counted to avoid leaving out trips taken by riders who make one or more transfers. Thus, if a rider is on their way to work and has to transfer from the bus to the Green Line to the Red Line, their commute would be counted as three rides. The data is completely anonymized, ensuring that no individual can be identified through the collected information. As a result, there are no privacy or ethical concerns regarding its collection, as it adheres to best practices for data security and confidentiality. However, there may be potential biases in the data due to the AFC system potentially providing inaccurate estimates when APC coverage is low, and in cases where passengers are manually counted, human error could affect the accuracy of the data.

Data Science Approaches

For our analysis, we employed a range of data science techniques to explore and model MBTA ridership patterns and trends. Key libraries utilized included Pandas and NumPy for data manipulation, Matplotlib and Seaborn for visualization, and Statsmodels and Scikit-learn for statistical modeling, machine learning, and forecasting.

We began each plot by using Pandas to read the ridership data from a CSV file into a DataFrame, enabling efficient data manipulation. Pandas was especially valuable for filtering the data by time period and grouping it by factors such as service line and day type, allowing us to segment the data and conduct targeted analyses. To visualize the data and highlight key trends, we used Matplotlib and Seaborn. These libraries were essential for transforming raw data stored in DataFrames into clear, insightful visualizations. With their flexibility and wide range of customization options, we presented our findings in a way that made complex patterns easier to interpret and derive meaningful conclusions from the data. A key aspect of our analysis was utilizing machine learning techniques we learned this semester and predictive modeling to uncover patterns and make forecasts from the ridership data. We applied K-means clustering from the Scikit-learn library, an unsupervised technique that grouped similar ridership patterns into clusters, helping us identify distinct behavioral trends over time. Additionally, we utilized STL (Seasonal-Trend decomposition using Loess) from the Statsmodels library, which decomposes time series data into overall trend and seasonal components for regression.

In summary, the tools and methods we employed helped us transform complex MBTA ridership data into well-structured insights, uncovering underlying trends and patterns that might have otherwise been overlooked. Together, these data science techniques provided valuable insights that deepened our understanding of the MBTA as a whole.

Results and Conclusions

Our initial visualization of monthly ridership by transit line from 2019 to 2024 revealed the significant impact of the COVID-19 pandemic, with ridership across all lines plummeting between February and April 2020. The graph also highlighted the relative popularity of each transit mode: buses led by a wide margin with over 6 million monthly rides as of December 2024, followed by the Red, Orange, and Green lines, as well as the Commuter Rail, each recording approximately 2–3 million monthly rides. In contrast, the Blue and Silver lines maintained lower ridership levels, around 1 million monthly rides. To simplify the visualization, all remaining lower-ridership services were grouped under an “Other” category. Importantly, the graph illustrates that ridership—while steadily recovering—remains below pre-pandemic levels, likely due to the continued prevalence of hybrid work and school models that have reduced the overall demand for public transit.

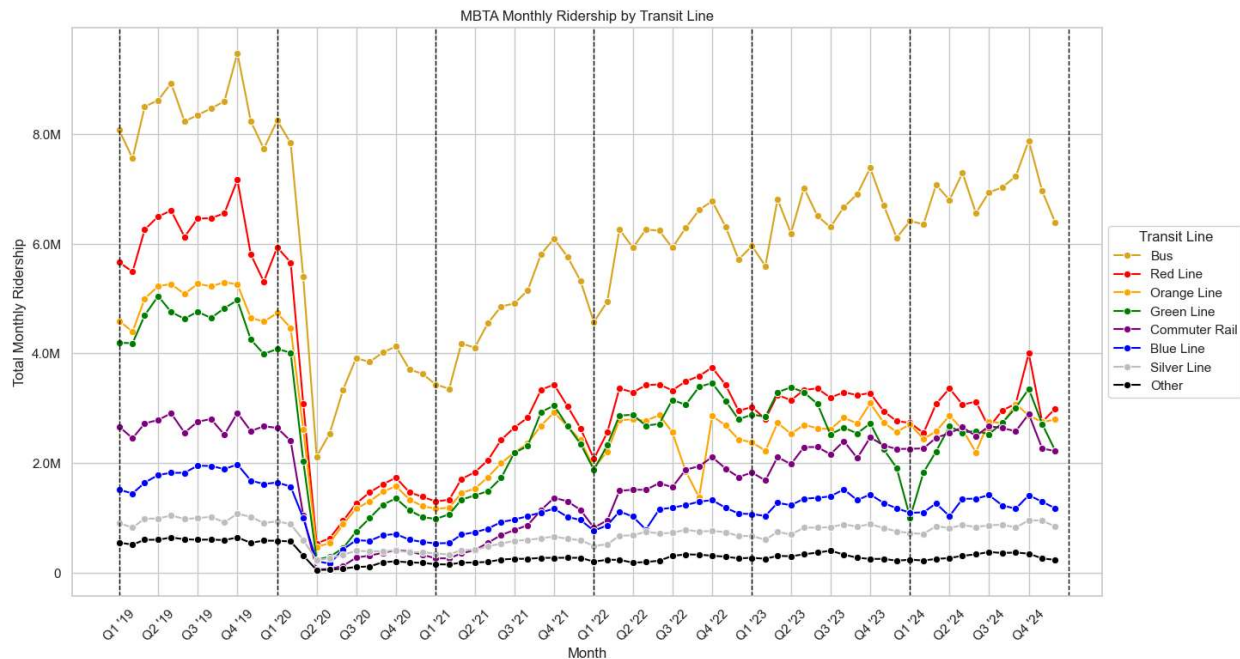


Figure 1: MBTA Monthly Ridership by Transit Line (2019-2024)

Next, we examined how daily ridership varies by service line and day type, using the MBTA's classification of days into three categories: Weekday, Saturday, and Sunday. Our second visualization features two bar graphs—one for 2019 (pre-COVID) and one for 2024—displaying average daily ridership by service line and highlighting the relative decline in weekend ridership. While overall ridership has decreased since 2019, as indicated by the smaller bars in the 2024 graph, the proportion of weekend ridership has increased, as shown by the growing percentage shares on the Saturday and Sunday bars. Notably, the Commuter Rail stands out with an absolute increase in weekend ridership, even as weekday usage has slightly declined. This shift may reflect increased recreational and leisure travel on weekends, as well as changes in commuting behavior driven by more flexible or remote work arrangements. As fewer people rely on transit for daily commuting, weekend usage may become a more prominent driver of overall ridership patterns.

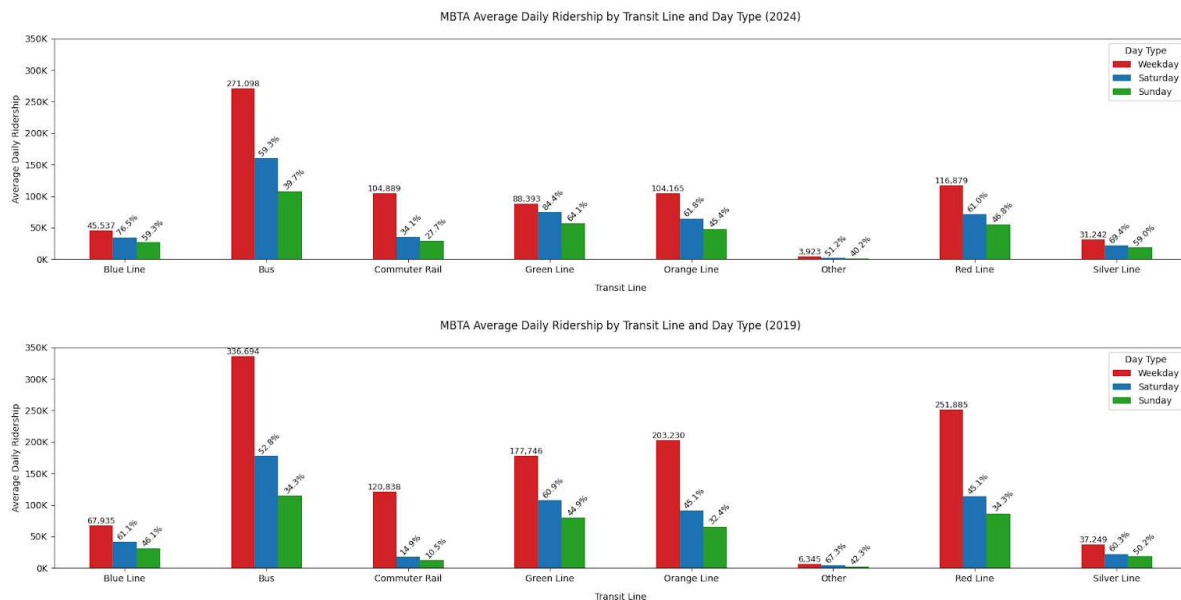


Figure 2: MBTA Average Daily Ridership by Transit Line and Day Type (2019 and 2024)

We then analyzed monthly ridership data from 2019 to 2024 side by side, revealing both long-term trends and consistent seasonal patterns. Ridership in 2019 was the highest by a substantial margin, regularly exceeding 30 million monthly rides during peak months. In contrast, 2020 experienced a dramatic collapse in ridership, plunging below 10 million monthly rides in the wake of the COVID-19 pandemic. In the years that followed, ridership gradually rebounded, though the pace of recovery slowed notably after 2021. Despite the pandemic's impact, seasonal fluctuations in ridership remained relatively stable across the dataset—excluding the outlier year of 2020—with ridership typically peaking in October and dipping to its lowest levels in February. These trends are likely influenced by a mix of weather-related factors and behavioral patterns: colder winter months tend to discourage public transit use, while fall often brings an uptick due to the return of school and regular work schedules under more favorable weather conditions.

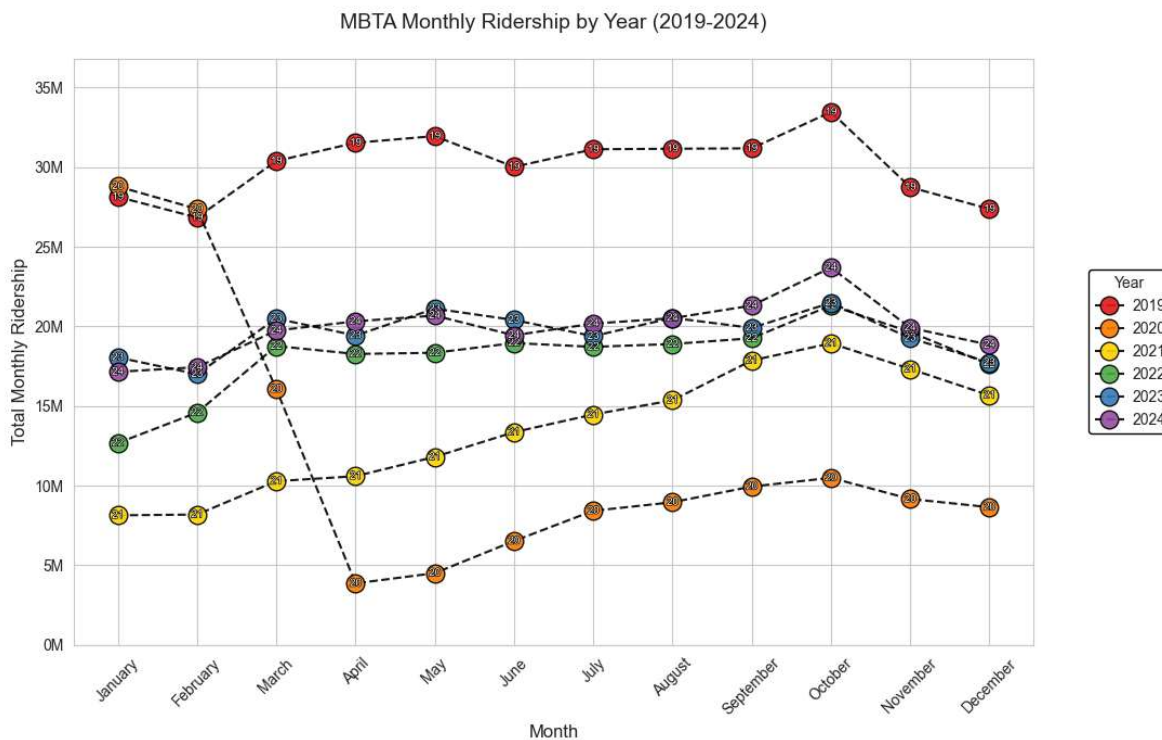


Figure 3: MBTA Monthly Ridership by Year (2019-2024)

Next, we applied machine learning techniques to explore hourly ridership patterns at the Northeastern University Green Line station, focusing on Fall 2024 data (as defined by the MBTA: August 25 to December 14, 2024). Recognizing the variation in ridership behavior across different types of days, we segmented the data into Weekdays, Saturdays, and Sundays. Our analysis centered on average net onboardings (boardings minus exits) by hour, separated into two points based on travel direction. To identify common patterns, we implemented k-means clustering with $k=3$, which revealed a morning cluster where directionality had minimal influence on net ridership, alongside more directionally distinct trends later in the day. Specifically, we observed that westbound trains experienced more exits, while eastbound trains saw higher onboardings during afternoon and evening hours. This asymmetry likely reflects Northeastern’s geographic position relative to downtown Boston, with the majority of Green Line stops—and key destinations—located to the east. As a result, commuter traffic tends to move eastward into the city and return westward later in the day.

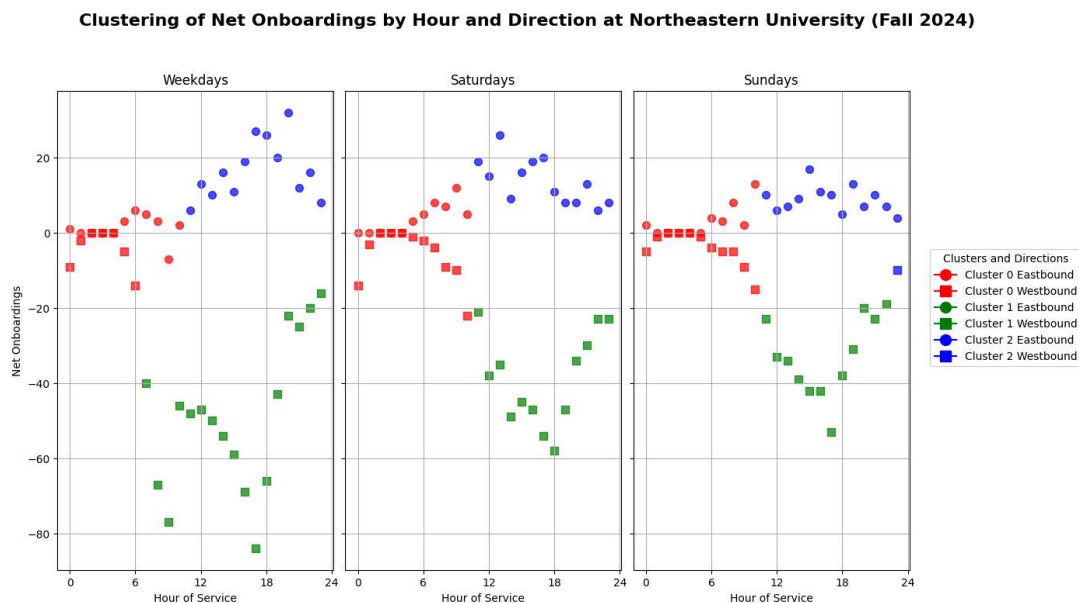


Figure 4: Clustering of Net Onboardings by Hour and Direction at Northeastern University (Fall 2024)

We then expanded our analysis to examine average hourly ridership trends across all stations and train lines. On weekdays, the data revealed two distinct peaks: a morning surge of just under 40 thousand boardings between 7–9 a.m., and a larger evening peak of approximately 46 thousand boardings between 4–6 p.m., corresponding with typical commuting hours. In contrast, weekend ridership displayed a more concentrated pattern, with a single afternoon peak—around 30 thousand boardings on Saturdays and 23 thousand on Sundays—occurring between 2–4 p.m., likely reflecting recreational travel. In addition to boardings, we analyzed net onboardings (boardings minus exits) over time to gain further insight into transit system dynamics. Positive net onboardings indicated periods when more people were entering than exiting the system, often during the buildup of commute periods. Negative net onboardings, conversely, signaled times when exits outpaced entries, reflecting the wind-down of travel activity. As expected, the net onboardings averaged zero across the day, consistent with the fact that all riders must eventually exit the system after entering.

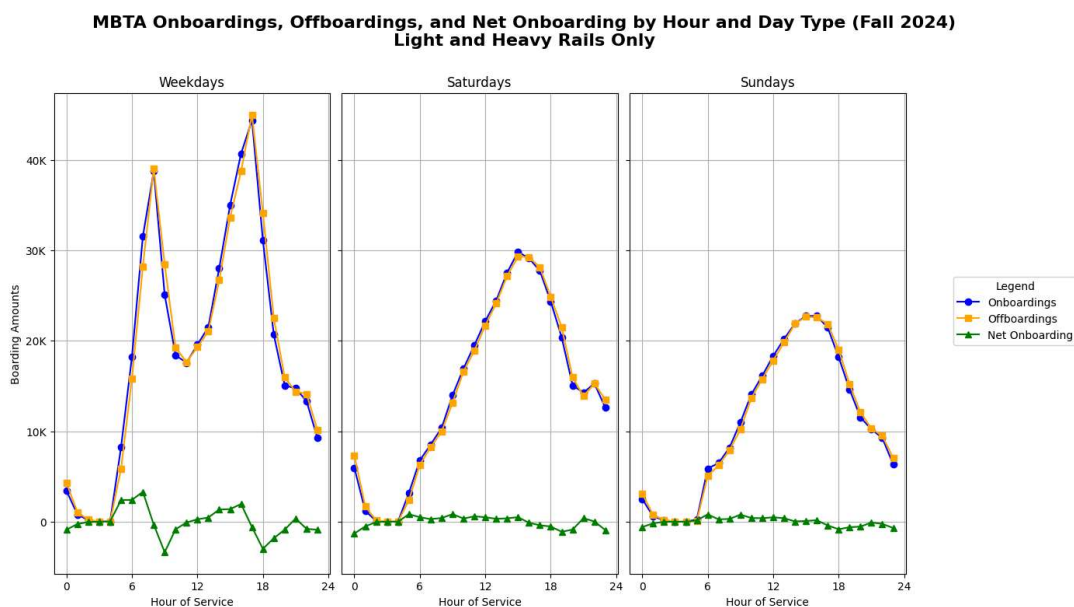


Figure 5: MBTA Onboardings, Offboardings, and Net Onboarding by Hour and Day Type (Fall 2024)

By combining the y-axes of these three graphs into a single, unified scale, we gain a clearer perspective on overall ridership trends. Combining day types in the analysis reveals a single dominant peak in both average onboardings and offboardings during the afternoon hours. However, it's important to consider that the dataset contains more weekdays than weekends, so the combined average is somewhat skewed. A properly weighted average would likely resemble the weekday pattern more closely, with its characteristic dual peaks. Notably, the average number of onboardings and offboardings remains nearly identical across all graphs—an expected result that reflects the natural balance of passengers entering and exiting the transit system.

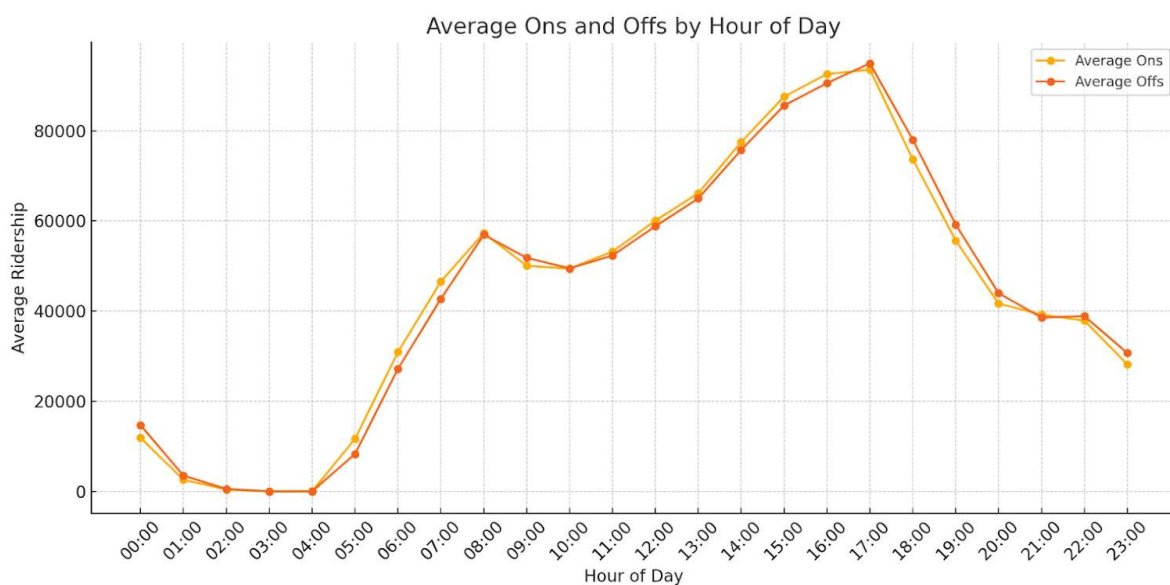


Figure 6: Average Ons and Offs by Hour of Day

Furthermore, when averaging all onboardings and offboardings across the dataset—spanning all hours, stations, service lines, and directions—the values were found to be identical. This outcome aligns with the aforementioned principle that every rider who enters the system must eventually exit. A deeper analysis by rail type revealed that heavy rail services

experience significantly higher ridership compared to light rail. This finding is consistent with our earlier observations regarding ridership trends by service line. The heavy rail network, which includes the Red, Orange, and Blue Lines, naturally accommodates a larger volume of passengers than the light rail system, which primarily consists of the Green Line and the less trafficked Mattapan Trolley.

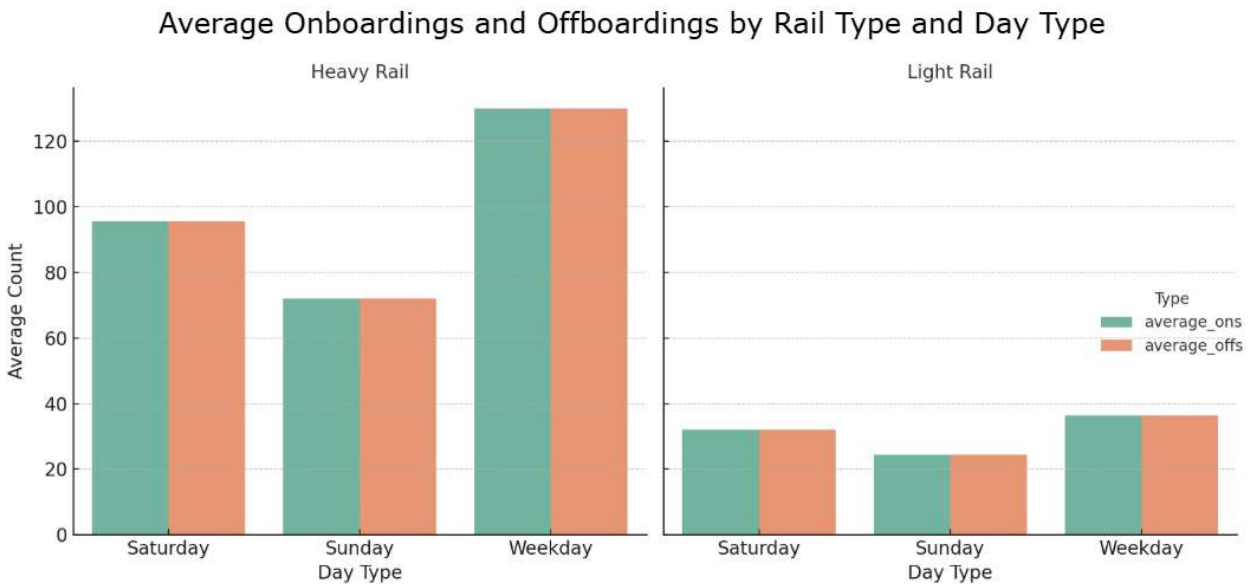


Figure 7: Average Onboardings and Offboardings by Rail Type and Day Type

Building on the monthly trends we previously analyzed, we applied predictive modeling to estimate ridership for 2025. Using an STL (Seasonal-Trend decomposition using Loess) model based on data from 2022 to 2024, we decomposed the ridership data into an overall trend and seasonal fluctuations. The model indicated a steady year-over-year increase in ridership, averaging just under 1 million additional monthly rides, suggesting that overall, we can expect continued growth through 2025. However, this growth is not expected to follow a linear path throughout the year. We anticipate typical seasonal variations, with ridership dipping in the early months, gradually rising through the spring and summer, and experiencing a sharp peak in the

fall. Our model forecasts that monthly ridership will peak in October, reaching approximately 24.3 million, marking the highest monthly ridership since February 2020—just before the onset of the COVID-19 pandemic.

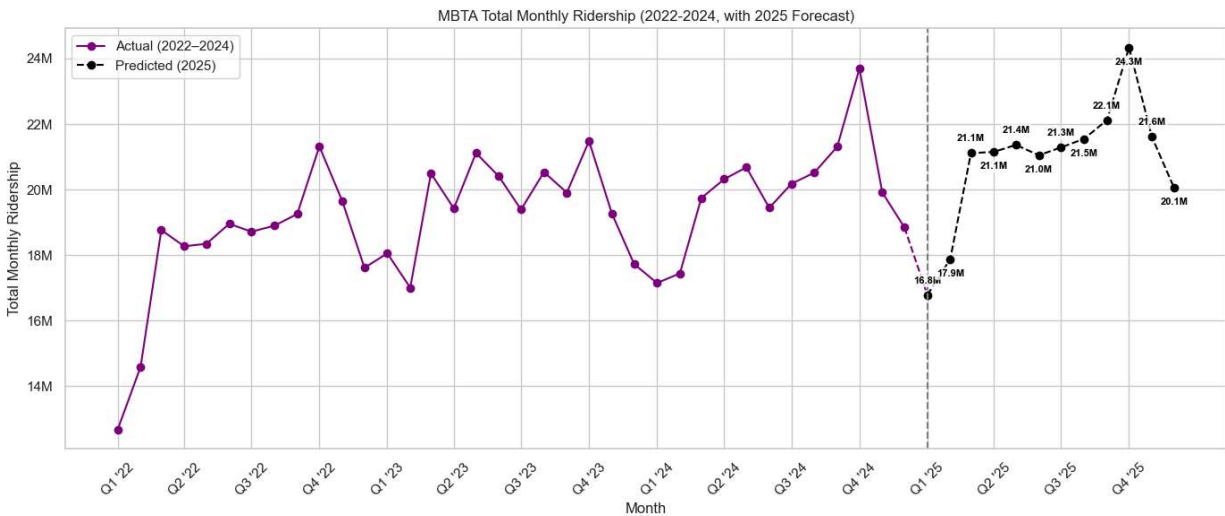


Figure 8: MBTA Total Monthly Ridership (2022-2024, with 2025 Forecast)

We then explored a new metric: average daily flow. Flow captures the total number of rides passing through a station, regardless of whether passengers are boarding or exiting at that station, providing a more complete picture of station activity. We visualized the top 10 stations by flow, including subtotals for stations that serve as intersections for multiple lines. As expected, the highest-ranking stations were all major transfer points, with Park Street and Downtown Crossing topping the list at over 250 thousand average daily riders in Fall 2024. Notably, all of the top 10 stations are located in central Boston, reflecting the high volume of movement throughout the downtown area for work, school, leisure, and more. These results reinforce the pattern of ridership density being concentrated in the core of the city.

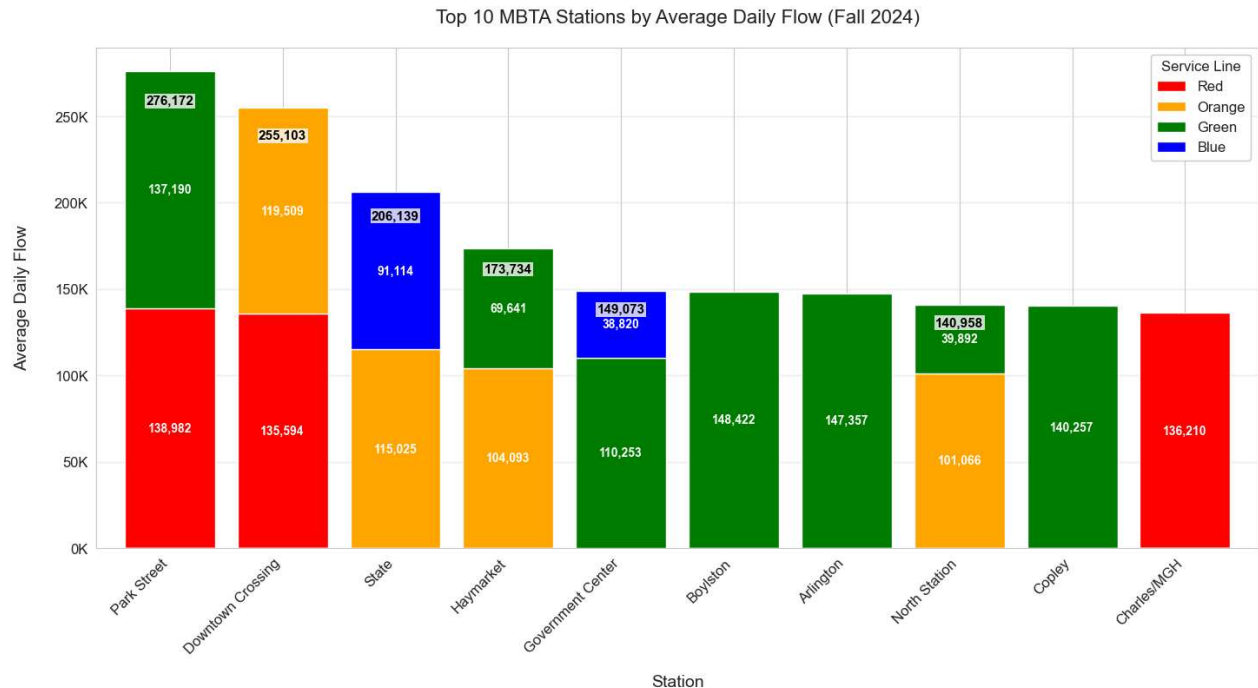


Figure 9: Top 10 MBTA Stations by Average Daily Flow (Fall 2024)

Lastly, to delve deeper into flow trends, we created a heatmap of the top 15 stations by average daily flow for Fall 2024, segmented by hour of the day. This visualization integrates insights from our earlier analysis of hourly ridership patterns with station-level flow data, offering a more comprehensive understanding of when and where passenger traffic is most concentrated. As expected, Park Street and Downtown Crossing show the highest ridership during the late afternoon and early evening hours, particularly between 3-6 p.m., with average hourly passenger flow surpassing 20 thousand for both stations during this period. These two central stations consistently maintain flow levels above 10 thousand throughout most of the day, highlighting their significance as key transit hubs. Other stations also experience elevated flow during peak periods, especially in the evening. Since this data aggregates both weekdays and weekends, afternoon ridership generally exceeds morning commute levels, although many stations still show a notable surge around 8 a.m. Overall, the stations with the highest peak-hour

traffic also rank the highest in total flow, further emphasizing their critical role as vital connectors within the MBTA network.

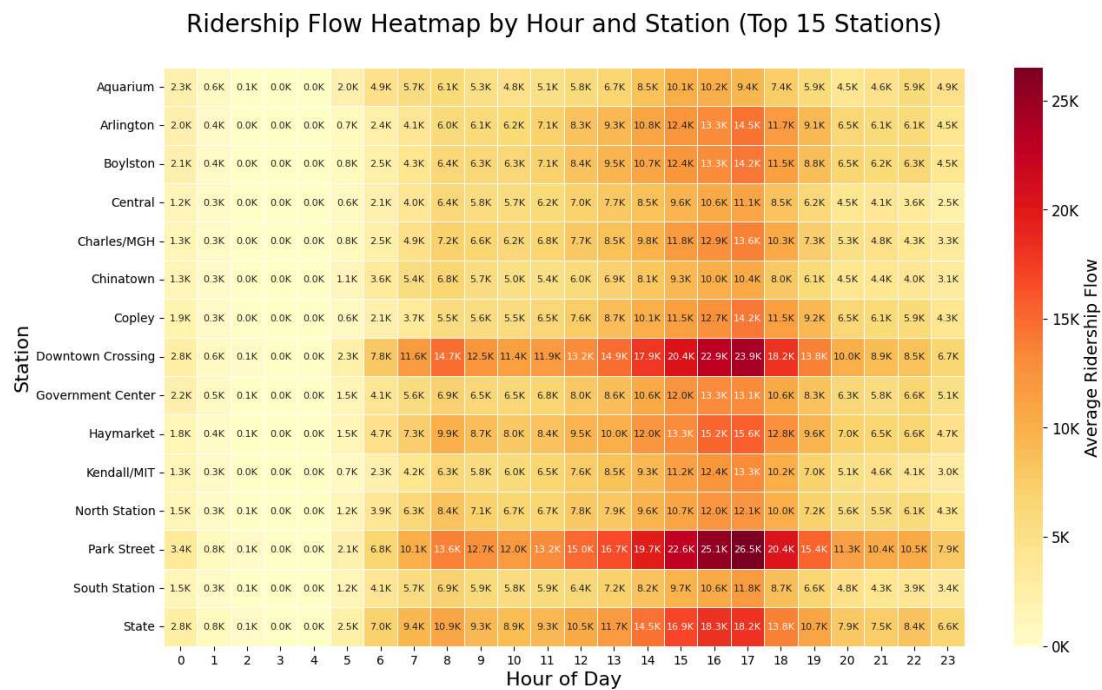


Figure 10: Ridership Flow Heatmap by Hour and Station (Top 15 Stations)

Future Work

In the future, beyond the scope of our project, we would aim to dig deeper into the ridership trends we uncovered, focusing particularly on anomalies, sharp declines, or persistent gaps across service lines and stops. We would also explore more targeted analyses, potentially identifying specific stations, times, or ridership groups that could benefit from adjustments in service. In this direction, we would seek out data with geographic aspects, allowing us to use GIS (Geographic Information System) tools like GeoPandas to create visual maps that depict performance across all service lines. These geographic visualizations would help highlight areas of opportunity or concern across all regions in which the MBTA operates, making it easier to understand regional patterns and spatial inequalities in service coverage.

Additionally, we would explore predictive modeling in greater depth, aiming to improve our forecasting and uncover insights that could guide MBTA decision-making. Our goal would shift from simply understanding trends as college students who rely on the system for our daily commutes to using our analysis to provide actionable insights that could influence the MBTA's operational decisions. By refining our techniques and focusing on data-driven improvements, we would aim to inform real-world changes that optimize transit services, enhance user experience, and better serve the Boston community as a whole.